Algorithm 1: $\theta_{k+1} \leftarrow \texttt{DistributedFixedRolloutUpdate}(\theta_k, ((\mathfrak{R}_g, \mathcal{D}_g, \mathcal{L}_g))_{g\in\mathcal{G}}, \mathsf{m})$

/* Targets, old-policy outputs, topology, and the empirical decision law stay frozen; only θ changes. */
/* In branched-RB, $\mathfrak{R}_g$ is the one rollout tree on GPU g; sequential records contain trajectory starts. */

Parameters: finite PBG; rollout profile $\pi = \pi^{\theta_k}$; $\mathsf{m} \in \{\mathsf{normal}, \mathsf{RB}, \mathsf{branched\text{-}RB}\}$; fixed GPU-local rollout outputs from `MultiGPUQueuedRollout`; GPU set $\mathcal{G}$

Parameters: execution-family priority poset $(\mathcal{M}, \leq_{\mathcal{M}})$ and public routing map κ; mode-correct per-decision loss $\texttt{ModeLoss}_{\mathsf{m}}$; batches are keyed only by $m \in \mathcal{M}$

Hyperparameters: K, common number of synchronous optimizer steps; calibrated family execution-cost curves $C_{g,m}(b)$ and admissible integer bounds/quanta for logical stratum sizes $B_{g,m}$, with $B_{g,m} \geq 1$ whenever the stratum has positive mass; common predicted per-GPU time budgets T_g; optimizer, common AMP loss scale, and global gradient-clipping rule; optional frozen-reference cache

Definitions: $\mathcal{D}_{g,m} \leftarrow \{\mathsf{s} \in \mathcal{D}_g : \kappa(\ell_g(\mathsf{s})) = m\}$; for a player-decision occurrence s,

$$\omega_g(\mathsf{s}) \leftarrow \begin{cases} 1, & \mathsf{m} \in \{\mathsf{normal}, \mathsf{RB}\}, \\ \displaystyle\prod_{\mathsf{r} \prec \mathsf{s}} \frac{1}{N_g(\mathsf{r})}, & \mathsf{m} = \mathsf{branched\text{-}RB} \end{cases}$$

Definitions: $Z_{g,m} \leftarrow \sum_{\mathsf{s}\in\mathcal{D}_{g,m}} \omega_g(\mathsf{s})$, $Z_g \leftarrow \sum_m Z_{g,m}$, $Z \leftarrow \sum_g Z_g$, and, when $Z_{g,m} > 0$,

$$q_{g,m}(\mathsf{s}) \leftarrow \frac{\omega_g(\mathsf{s})}{Z_{g,m}}, \qquad \alpha_{g,m} \leftarrow \frac{Z_{g,m}}{Z}.$$

Definitions: $\texttt{ModeLoss}_{\mathsf{m}}(\theta; \mathsf{s}, \mathcal{O}_g)$ is the corresponding loss of `core_math`: posterior sums are inside the RB losses and, in branched-RB,

$$\texttt{ModeLoss}_{\mathsf{branched\text{-}RB}}(\theta; \mathsf{s}, \mathcal{O}_g) = \frac{1}{N_g(\mathsf{s})} \sum_{\mathsf{a}\in\mathfrak{C}_{\mathsf{s}}} \mathcal{L}^{\pi}_{g,\mathsf{s},\mathsf{a}}(\theta) + \mathcal{L}^{\mathsf{node}}_{g,\mathsf{s}}(\theta),$$

where $\mathcal{L}^{\pi}$ includes the edge target and $\widehat{\mathbb{P}}(x_p \mid \mathsf{sa})$ weight, while $\mathcal{L}^{\mathsf{node}}$ contains the enabled value, entropy, and reference-KL terms with $\widehat{\mathbb{P}}(x_p \mid \mathsf{s})$ weight. The rollout intensity β is an external black box and has no head, target, or loss in this algorithm

Definitions: $\widehat{\sigma}^2_{g,m}$ is a frozen lagged estimate of $\mathrm{trCov}_{q_{g,m}}[\nabla_\theta \texttt{ModeLoss}_{\mathsf{m}}]$, or a configured cheap proxy calibrated against it; it affects only learner work allocation, never the objective or rollout law

Output: identical updated parameters θ_{k+1} and optimizer state on every GPU

foreach $g \in \mathcal{G}$ ***as soon as its local rollout completes*** **do**
  Compute $(\omega_g(\mathsf{s}))_{\mathsf{s}\in\mathcal{D}_g}$ and $(Z_{g,m})_{m\in\mathcal{M}}$ in one pass over the local rollout records
end
$Z^\star \leftarrow \texttt{IAllReduceSum}_g(Z_g)$ without a preceding barrier
foreach $g \in \mathcal{G}$ ***in parallel*** **do**
  Stream terminal rewards or pre-terminal completion microbatches through the mode-correct backward reduction frontier and construct the fixed compact struct-of-arrays target table $\mathcal{O}_g \leftarrow \texttt{BackwardTargets}_{\mathsf{m}}(\mathfrak{R}_g, \mathcal{D}_g, \mathcal{L}_g; V^{\theta_k})$, laid out by m but never by street
  /* completed child summaries are contracted immediately */
  foreach $m \in \mathcal{M} : Z_{g,m} > 0$ **do**
    $Q_{g,m} \leftarrow \texttt{Alias}((\omega_g(\mathsf{s}))_{\mathsf{s}\in\mathcal{D}_{g,m}})$
  end
  If reference KL is enabled and its byte plan fits, evaluate the frozen reference once per distinct retained input, in family-only calls, and attach the exact sufficient reference outputs to $\mathcal{O}_g$; otherwise retain the inputs needed to recompute those exact outputs inside each physical family call
  Release leaf reaches, completion workspaces, reduction-frontier summaries, frozen-rollout transients, and topology not owned by the compact target table
end
$Z \leftarrow \texttt{Await}(Z^\star)$ and assert $Z > 0$
Choose integer $B_{g,m}$ under a common per-GPU time budget by minimizing the lagged variance proxy

$$\sum_{m:Z_{g,m}>0} \frac{(Z_{g,m}/Z)^2 \widehat{\sigma}^2_{g,m}}{B_{g,m}} \quad \text{subject to} \quad \sum_m C_{g,m}(B_{g,m}) \leq T_g.$$

Use $B_{g,m} \propto Z_{g,m}\widehat{\sigma}_{g,m}/\sqrt{c_{g,m}}$, where $c_{g,m}$ is the calibrated local linear slope of $C_{g,m}$, only as the linear-cost warm start; then round/search over efficient family batch quanta
$\theta \leftarrow \theta_k$ on every GPU
for $t = 0, \ldots, K-1$ **do**
  foreach $g \in \mathcal{G}$ ***in parallel*** **do**
    foreach $m \in \mathcal{M} : Z_{g,m} > 0$ **do**
      Sample $(\mathsf{S}_{g,m,b})_{b=1}^{B_{g,m}} \overset{\text{iid}}{\sim} q_{g,m}$ using $Q_{g,m}$
    end
    Set the rank-local stratified loss

$$\widehat{l}_{g,t}(\theta) \leftarrow \sum_{m:Z_{g,m}>0} \alpha_{g,m} \frac{1}{B_{g,m}} \sum_{b=1}^{B_{g,m}} \texttt{ModeLoss}_{\mathsf{m}}(\theta; \mathsf{S}_{g,m,b}, \mathcal{O}_g).$$

    Keep $\mathcal{O}_g$ GPU-resident and gather it only by device index tensors. Group the sampled logical decisions only by m, coalesce equal current-model inputs while retaining their loss multiplicities, pad and mask mixed public stages inside maximum-feasible family-m physical forward/backward microbatches without subdividing by street, and accumulate the exactly scaled physical loss sums into $G_{g,t} \leftarrow \nabla_\theta \widehat{l}_{g,t}(\theta)$ without an optimizer step; never average a physical microbatch independently
    Flatten the common-scale fp32 gradients in fixed parameter order into one preallocated buffer, inserting zeros for absent parameters
  end
  $G_t \leftarrow \texttt{AllReduceSum}_g(G_{g,t})$ on the flat buffers, with no division by $|\mathcal{G}|$
  Derive the identical nonfinite decision locally from the reduced flat buffer (no second collective), unscale once, apply the common global clipping rule, and set $\theta \leftarrow \texttt{OptimizerStep}(\theta, G_t)$ and the next loss scale identically on every GPU
end
$\theta_{k+1} \leftarrow \theta$
/* No path weight is multiplied into a sampled loss: it is already represented by the alias law and $\alpha_{g,m}$. */
/* Conditionally on the fixed distributed rollout, for every current θ,

$$\mathbb{E}(G_t \mid (\mathfrak{R}_g)_g, \theta) = \nabla_\theta \sum_{g\in\mathcal{G}} \sum_{\mathsf{s}\in\mathcal{D}_g} \frac{\omega_g(\mathsf{s})}{Z} \texttt{ModeLoss}_{\mathsf{m}}(\theta; \mathsf{s}, \mathcal{O}_g).$$

/* The flat SUM reduction is intentional: $\alpha_{g,m} = Z_{g,m}/Z$ already supplies the exact global normalization, so neither rank-local means nor DDP's default equal-rank averaging may be introduced. */
return θ_{k+1}